

RDK S600 NVMe SSD Expansion Tutorial

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The SSD included with the Yahboom kit has been compressed in capacity to facilitate flashing, resulting in an image of less than 100GB.

In this tutorial, we will demonstrate how to use a virtual machine to expand the SSD to its original capacity.

Preparing the Virtual Machine

You need to prepare a computer running Ubuntu, or install a virtual machine with Ubuntu installed.

Additionally, you will need an SSD enclosure to connect the SSD to your computer. (In theory, you can also directly insert it into the RDK S600 to begin operations, but this tutorial uses a virtual machine for demonstration.)

Installing GParted in the Virtual Machine

Open the terminal in Ubuntu.

Enter the following commands in the terminal:

```
sudo apt update
sudo apt install gparted -y
```

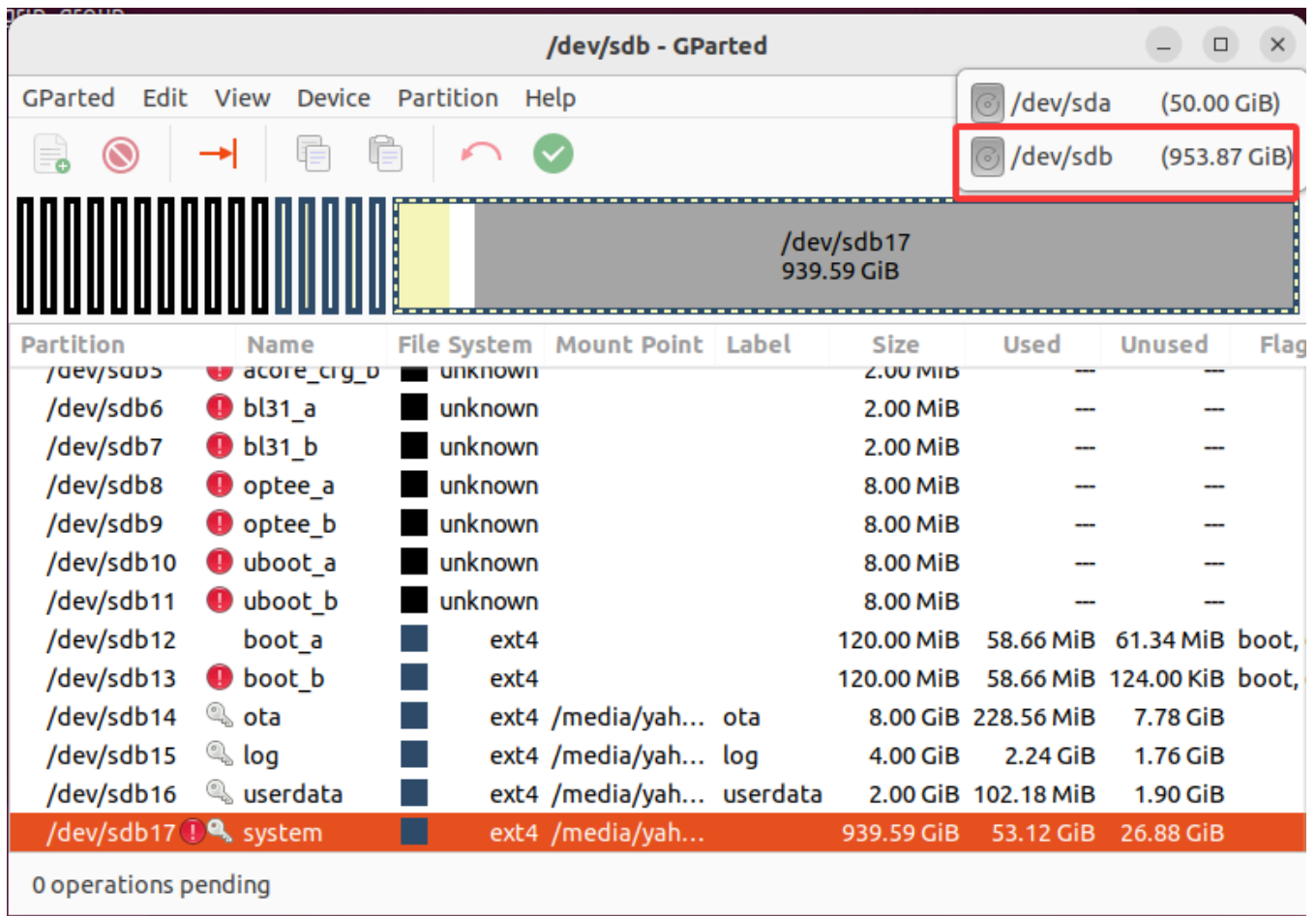
```
jetson@yahboom: ~  
jetson@yahboom:~$ sudo apt update  
[sudo] password for jetson:  
Hit:1 http://ports.ubuntu.com/ubuntu-ports jammy InRelease  
Hit:2 https://repo.download.nvidia.cn/jetson/common r36.4 InRelease  
Hit:3 https://repo.download.nvidia.cn/jetson/t234 r36.4 InRelease  
Hit:4 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease  
Hit:5 https://repo.download.nvidia.cn/jetson/ffmpeg r36.4 InRelease  
Hit:6 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease  
Hit:7 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
203 packages can be upgraded. Run 'apt list --upgradable' to see them.  
jetson@yahboom:~$ sudo apt install gparted -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  gparted-common  
Suggested packages:  
  dmraid gpart jfsutils kpartx mtools reiser4progs reiserfsprogs udftools xfsprogs exfatprogs  
The following NEW packages will be installed:  
  gparted gparted-common  
0 upgraded, 2 newly installed, 0 to remove and 203 not upgraded.  
Need to get 454 kB of archives.  
After this operation, 1,997 kB of additional disk space will be used.  
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 gparted-common all 1.3.1-1ubuntu1 [71.9 kB]  
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 gparted arm64 1.3.1-1ubuntu1 [382 kB]  
Fetched 454 kB in 3s (138 kB/s)  
debconf: delaying package configuration, since apt-utils is not installed
```

Connect the SSD to your computer, then launch GParted with sudo privileges.

Enter the following in the terminal:

```
sudo gparted
```

After launching GParted, select the correct drive in the upper-right corner (mine is `/dev/sdb`, using a 1TB SSD for demonstration).



Expanding the Partition

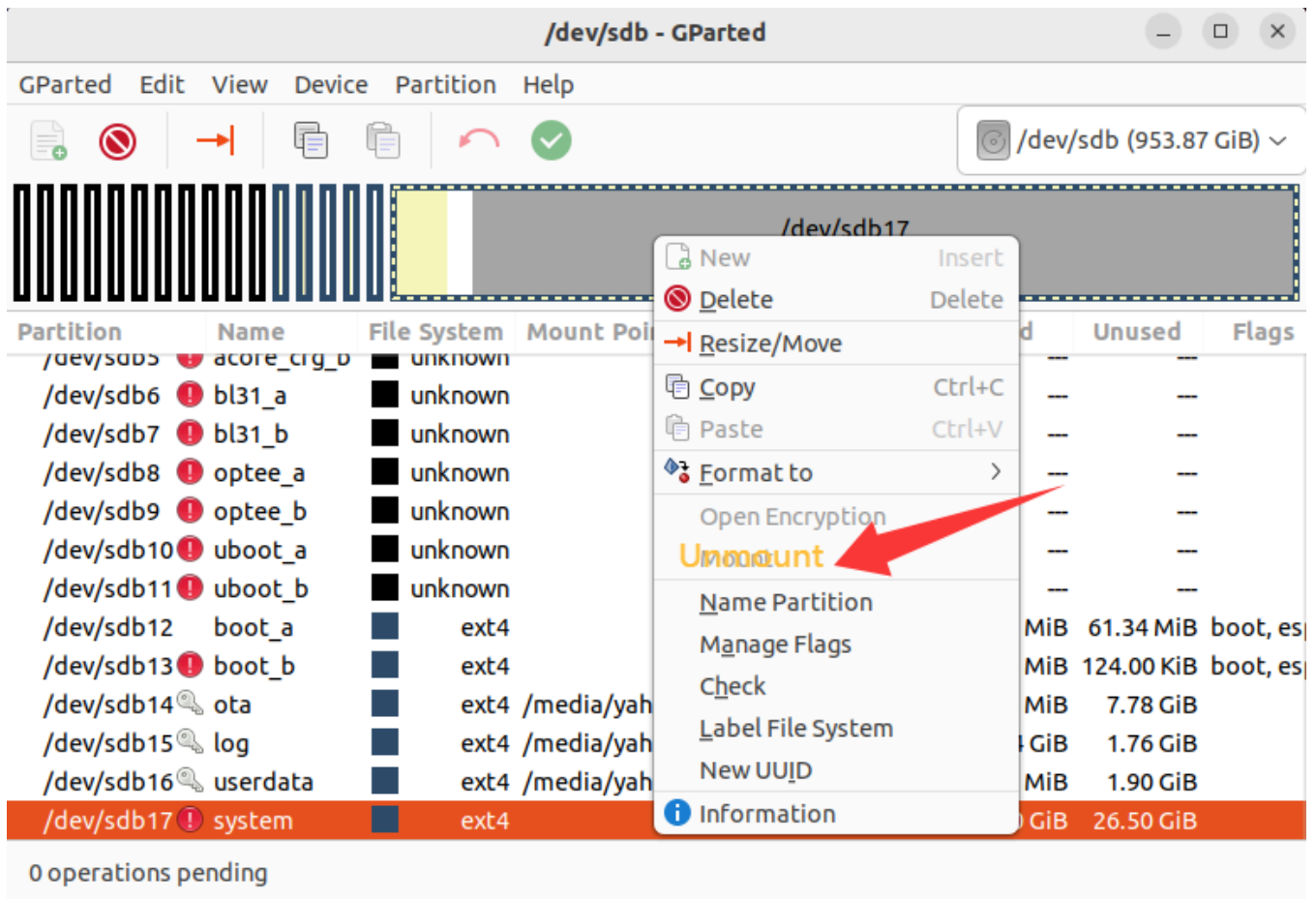
Right-click on the white partition at the top of GParted and click "Resize/Move":

There are **two scenarios**. Let's start with the first one.

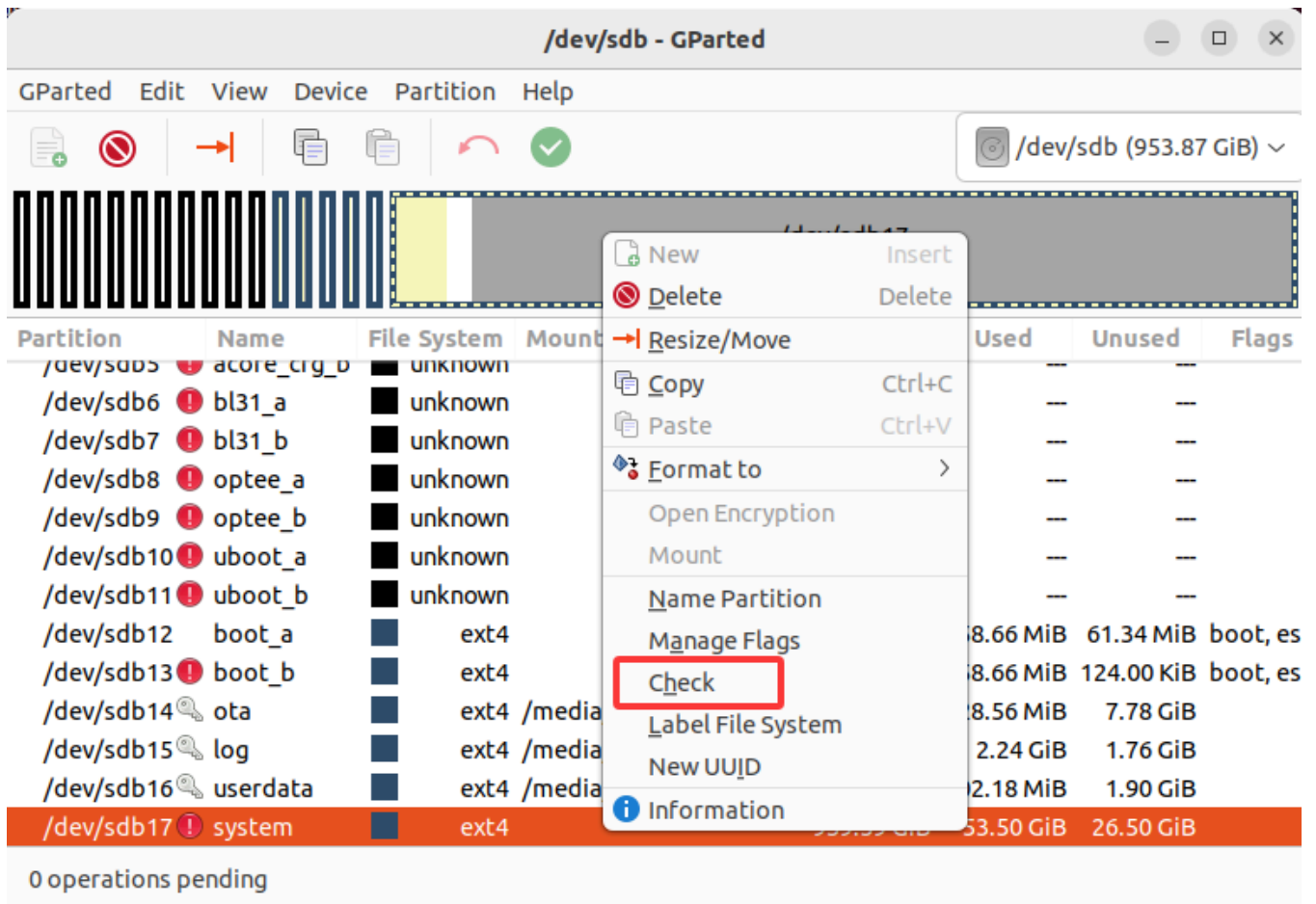
Scenario 1:

For a 1TB SSD that has been inserted into the RDK S600 and had the system started,

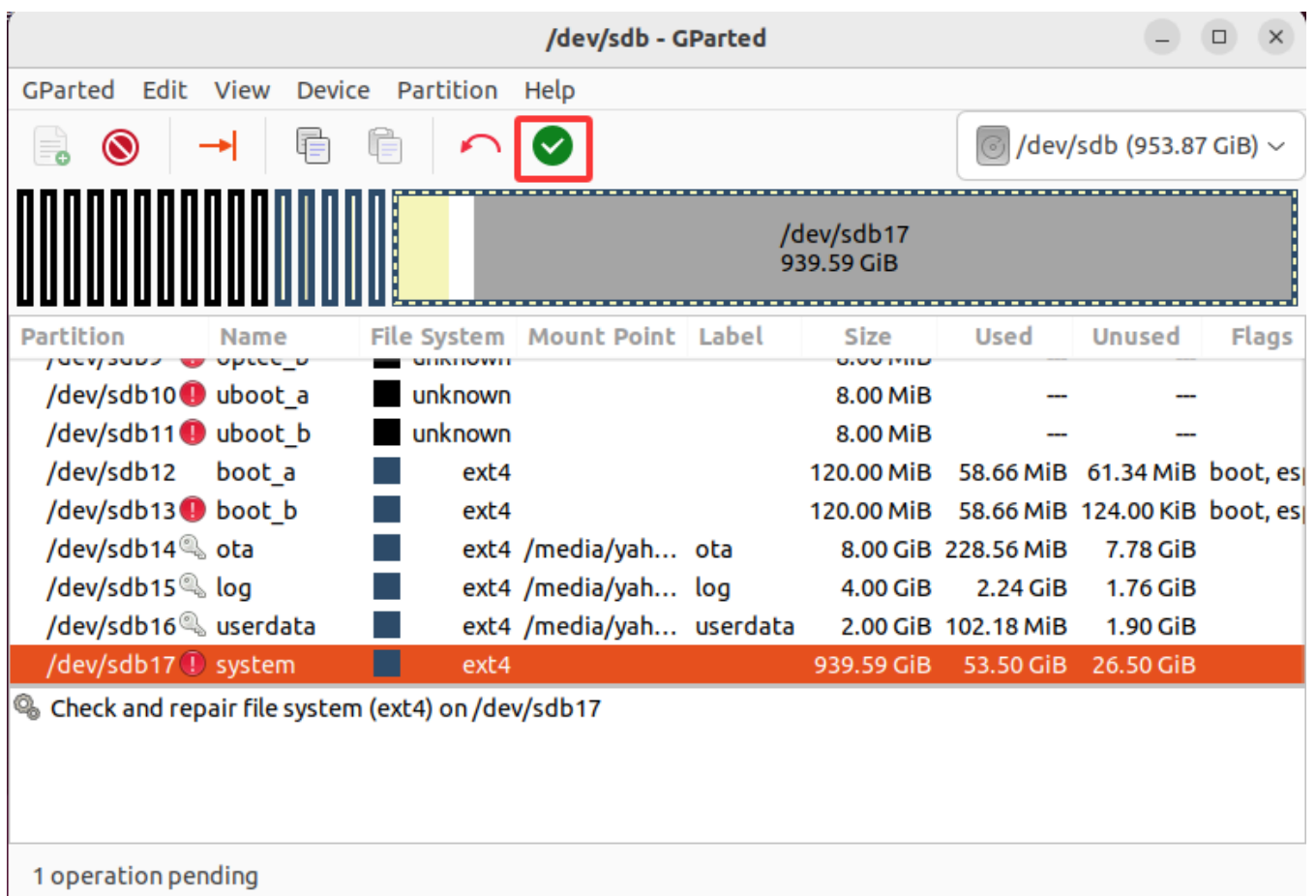
The partition display in GParted will look like this, with the yellow-white and gray areas merged together. We need to select this partition and click the Unmount button.



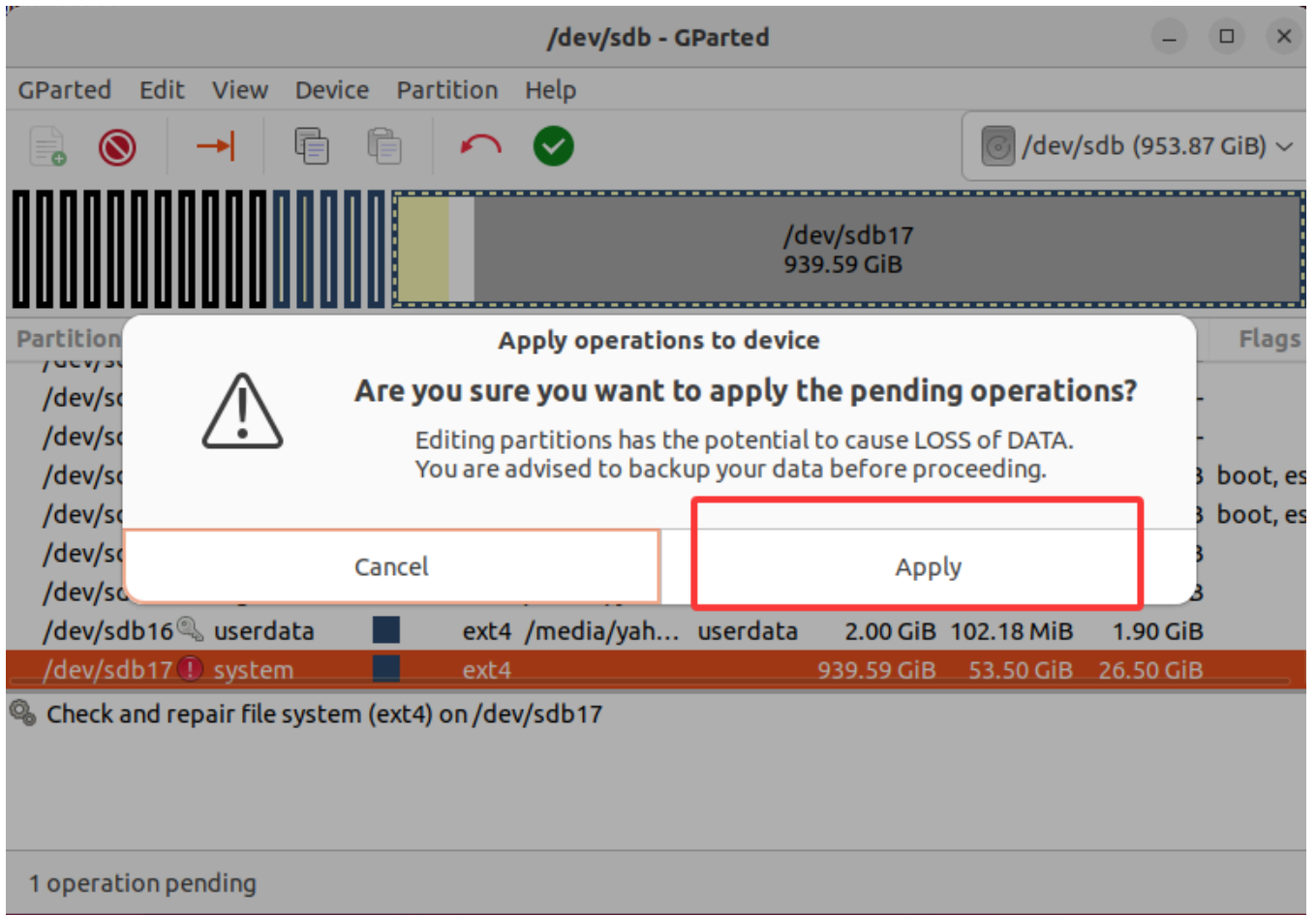
Then click Check.



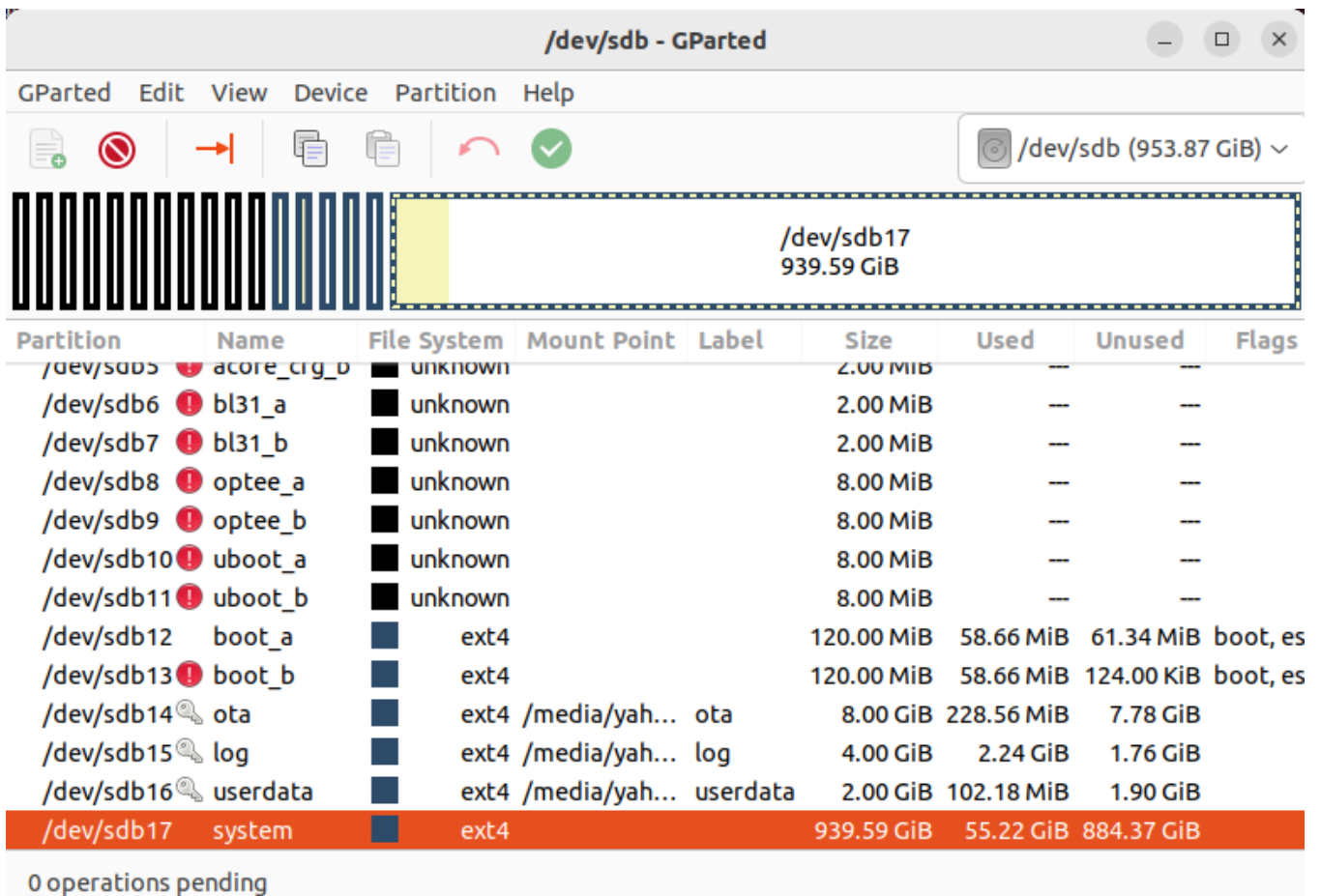
After Check completes, click the green checkmark.



Click Apply.

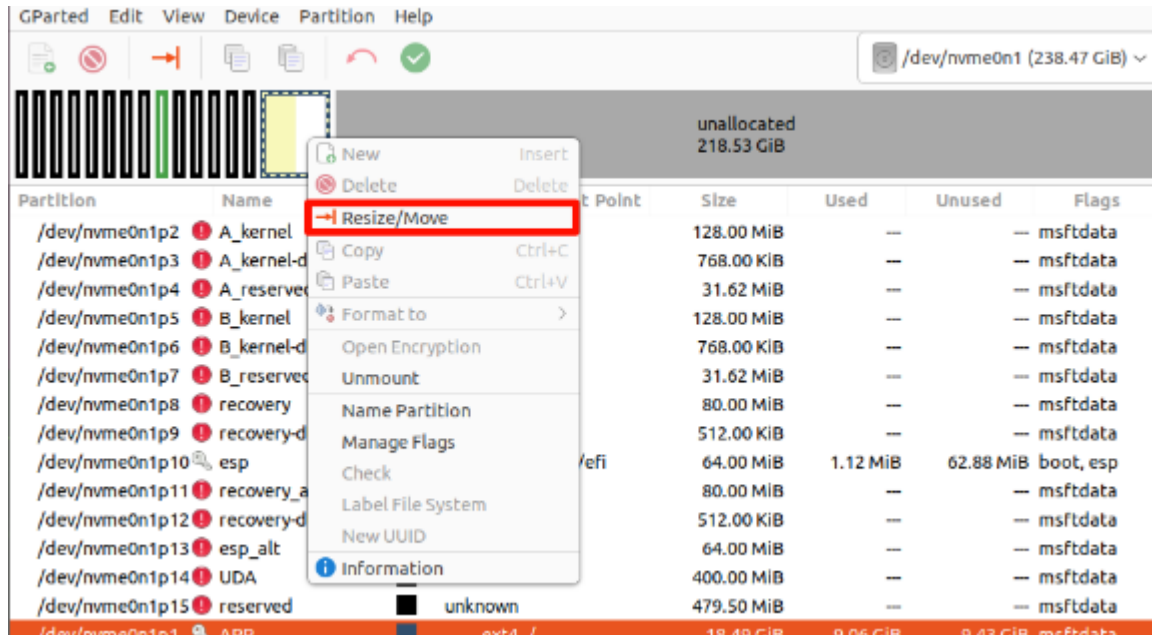


After waiting for some time, the expansion is successful.

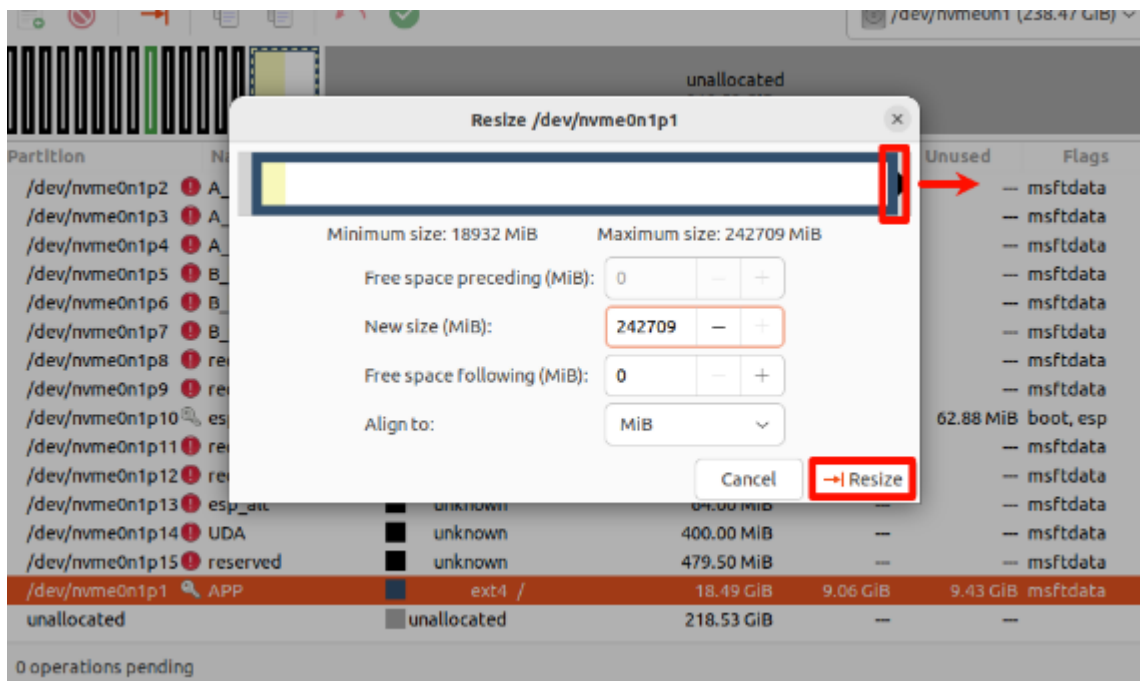


Scenario 2

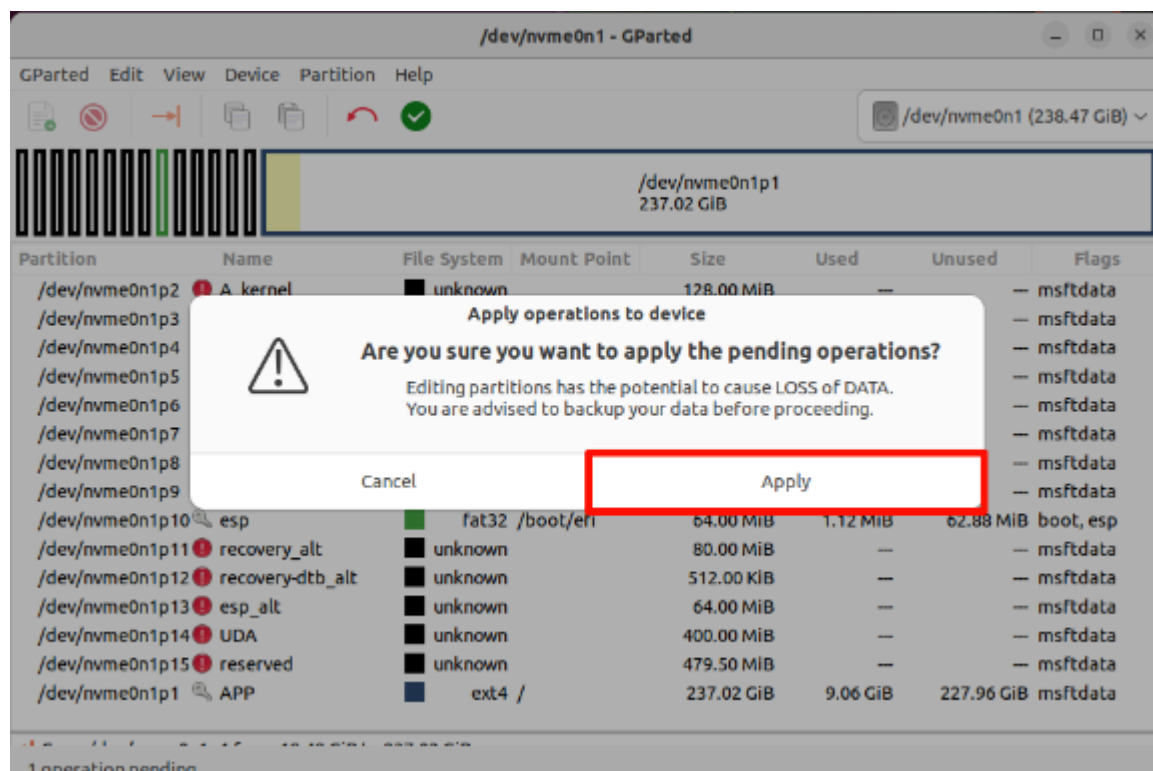
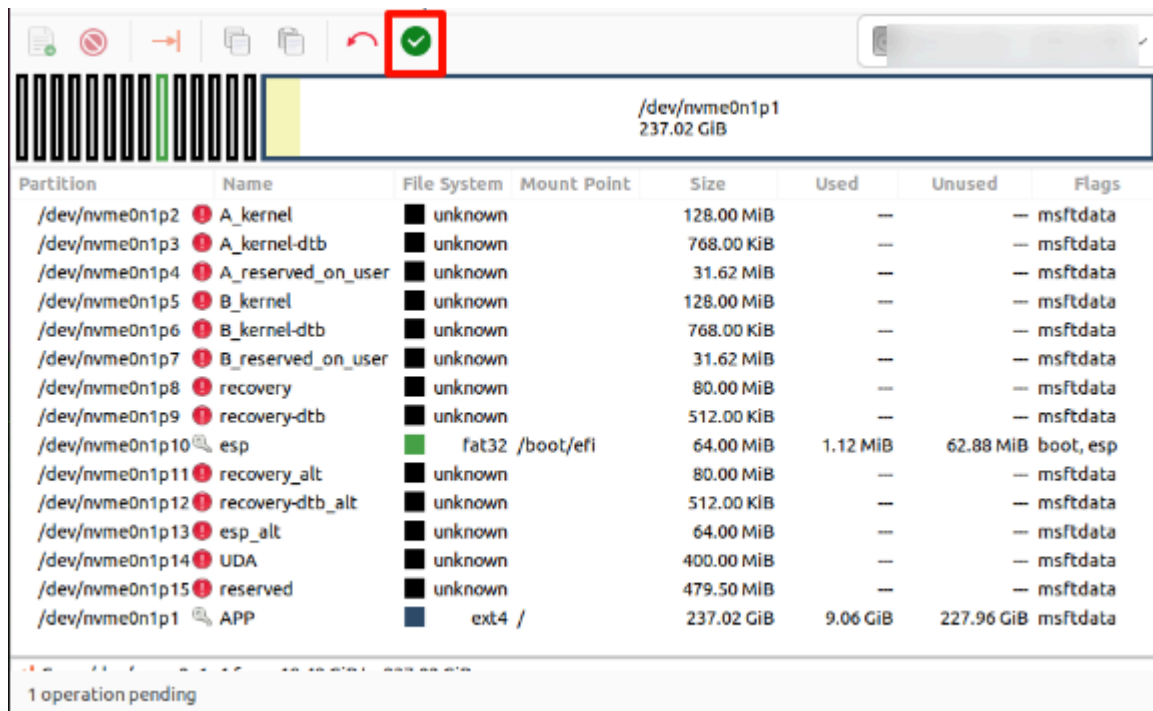
For 2TB SSDs or 1TB SSDs that have not yet been used with the RDK S600 motherboard, the partition in GParted will look like this (the gray area and yellow-white area are separate). In this case, make sure to click on the yellow-white partition, not the gray section behind it.



Click the Resize/Move button, drag the slider all the way to the right, then click Resize in the bottom-right corner.



Click the green confirmation button at the top.



After clicking Apply, the expansion will take some time.

